Database systems for big data

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Course Objectives

- Learn about the modern database system architecture and practices in database internals for analytical workloads.
- Students will learn about:
  - State-of-the-art architectures of a modern DBMS.
  - State-of-the-art research topics in a modern analytical DBMS.
  - Common practices in the implementation of DBMS components.
  - Examples of recent analytical DBMSs.

Course Objectives

- The focus this year will be on **query processing** in analytical database systems.
- Most approaches use existing technology designed for classical relational systems that is
  - Scaled to new hardware, faster networks, MPP, shared-nothing systems, shared disks, etc.
  - Examples:
    - Implementation of Hash join on multiple processors and cores.
    - Implementation of Sort-Merge join in SIMD environment (vectorization).

Course Literature

- Based on the course:
  - Advanced Database Systems, 2023 and 2024
  - Carnegie Mellon University (CMU)
  - Prof Andy Pavlo
- Each lecture: 2-3 papers.
  - The **first paper** is the mandatory reading per lecture.
- Transparencies (Prof. Andy Pavlo)
- Textbook from previous years:
  - Tamer Özsu, Patrick Valduriez, Principles of Distributed Database Systems, 4th Edition, Springer, ISBN 978-1-4419-8833-1, 2020.

Background

- I assume that you have already taken an intro course on databases.
  - Introduction to database systems (see e-classoroom)
- We will discuss modern variations of classical algorithms that are designed for today’s hardware.
- Things that we will not cover:
  - SQL, Relational Algebra, Storage Models, Memory Management, Basic Join Algorithms, Indexes (Hash, B-tree), Transaction management, Concurrency control.
  - Basic Algorithms, Data Structures, Sorting Algorithms, Hashing, etc.

Office hours

- After class in my office:
  - Thursdays, after the lecture.
  - Kettejeva 1, Compuer Science Department.
- Things that we can talk about:
  - Issues on seminar work.
  - Paper clarifications/discussion.
  - Any other topic related to the course.

Course Discussions

- Discussion through **Forum Discussions** in the e-classroom of the course:
  - https://e.famnit.upr.si/mod/forum/view.php?id=118168
- If you have a **technical question** about the seminars, please use forum Discussions.
  - Don’t email me directly.
  - In most cases, your question is of interest to other students.
- Other questions should be sent to me.

Topics

- Modern Analytical Database Systems
- Data Formats, Encoding, Compression
- Query Execution & Processing
- Vectorized Execution and Compilation
- Parallel Join Algorithms
- Scheduling and Coordination
- Query Optimization
- Modern System Analysis
  - BigQuery/Dremel, Snowflake, DuckDB, MongoDB
  - 23/24: DynamoDB, Bigtable, Spanner & F1

Reading Assignment

- One mandatory reading per class. You can skip two readings during the semester.
  - You have to read the paper! After this you prepare the report.
- You must submit a synopsis before each lecture:
  - Overview of the main idea presented (3 sentences).
  - Key findings/takeaways (2-3 sentences).
  - Description of the system discussed and how it was modified/extended (1 sentence).
  - Workloads / benchmarks used in evaluation (1 sentence).

Lecture Notes

- Each student will be assigned one class during the semester to write notes about the lecture's contents.
  - Summarize the key topics and material from the lecture.
  - Do not need to include ancillary discussions or when "Somebody goes off the chain".
- Notes will be available in e-classroom.
  - Samples are available in previous e-classrooms.
  - Must include paper citations (Bibtex).
  - You are allowed to use images from course slides.

Lecture Notes

- Each student must submit their notes as a PR within one week (seven days) of the lecture.
  - See list of students with assigned lectures in e-classroom for your assigned lecture date.
  - You are allowed to swap without notifying instructors.
- You are allowed to use AI to assist with this.
  - Please include information about what tools you used to help future students.
  - You are responsible for the contents of the notes.

Seminar

- Analysis of a prototype or production version of a modern analytical/transactional database management system.
  - I will provide a list of interesting database systems where you can choose the topic.
  - Selected database system has to be paired with an appropriate research paper (SIGMOD, VLDB, CIDR, etc.)
- Analysis of a novel DBMS
  - Research paper
  - Test application
  - Report
  - Presentation

Seminar

1. Select a DBMS among the systems proposed. Install the system on your local network (>=2).
2. Study the architecture and components of the system.
   - *Components*: storage model, txn management, query execution, query optimization, join algorithms, etc.
3. Define a simple application that uses the main features of the selected system.

- Make a system design.
- Implement the application.
- Experiment with the system.

Seminar

4. Describe the architecture, selected architectural element, and the implemented application in a short paper (6-10 pages).
5. Give a short presentation (10-15min) of the work to the class.

Grade Brakedown

- Seminar = 40%
- Written Exam = 30%
  - 90-120 min, 4-5 exercises
- Reading Assignement = 20%
- Lecture Notes = 10%

All parts must be positive, i.e., the grade have to be greater than 50%.

Plagiarism

- You may not do the following:
  - Copy text from a research paper or other sources that you find on the web in your seminar or reading assignment.
    - Each review must be your own writing.
  - Use the (updated) text from a colleague.
  - Use AI tools in case this is not explicitly allowed.
- Plagiarism will not be tolerated!
  - UP University rules apply.
  - Do not take a risk!